

Training Dynamics of Overparameterized Neural Networks

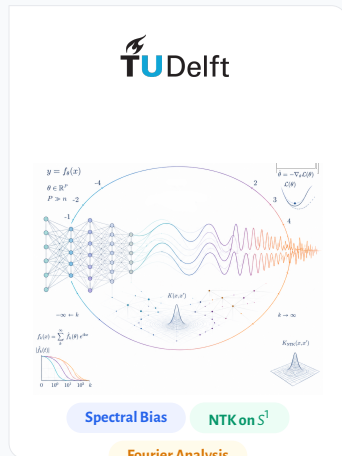
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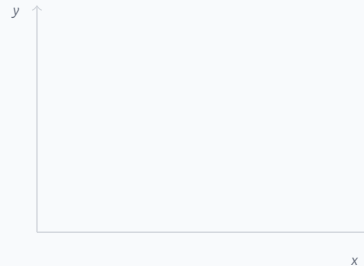
Advisor: Prof. Dr. D. M. J. Tax, Pattern Recognition & Bioinformatics Group

Supervisor: Prof. Dr. A. Heinlein, Numerical Analysis Group

Committee Member: Prof. Dr. A. Papapantoleon, Applied Probability Group



Supervised Learning: Fitting data to a function



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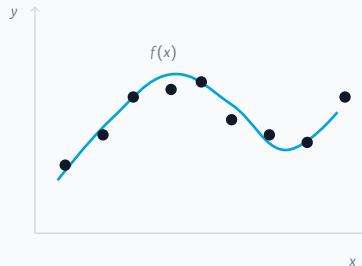


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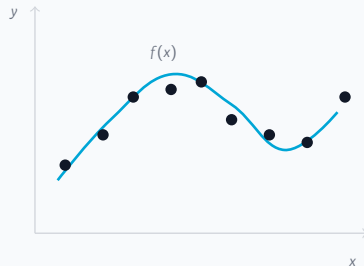
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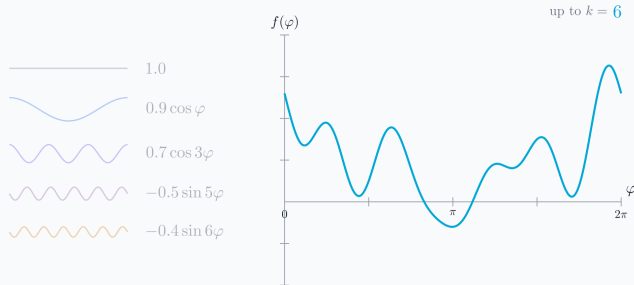
How: choose f so its predictions are close to the target function on

the observed data, i.e. minimise $\frac{1}{n} \sum_{i=1}^n (f(x_i) - f^*(x_i))^2$.



Example: a concrete target

A function is built from a coarse shape plus finer and finer detail.



Frequency components

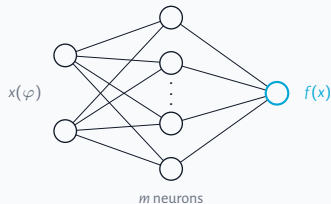
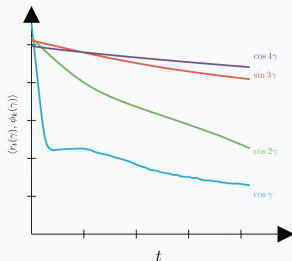
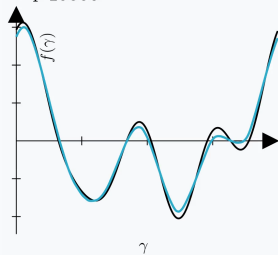
- ▶ Each piece is one frequency, from slow waves to fast ones.
- ▶ Low frequencies set the coarse shape; high frequencies add the fine detail.
- ▶ The target is a short sum of these pieces.

How do we learn this function?

Watch one network learn it

Train a single-hidden-layer network on that target and watch the residual — what gets learned first?

step 10000



$$f_m(x; \theta) = \frac{1}{\sqrt{m}} \sum_{i=1}^m a_i \sigma(w_i^\top x + b_i), \quad \sigma = \text{ReLU}$$

What to watch

- ▶ The fit climbs toward the target.
- ▶ Low-frequency modes die first.
- ▶ Fine detail fills in last.

The coarse shape appears first; the fine detail comes last.

Why does gradient descent learn simple structure first?

A spectral view of training dynamics

Thesis in one sentence

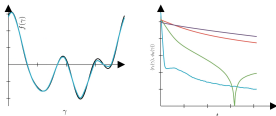
Reformulate spectral bias as an operator eigenvalue problem, then quantify how the picture changes at finite sample size and finite network width.

Three questions

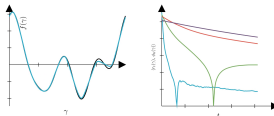
- 1 What controls mode-wise learning rates?
- 2 Does the explanation survive finite data?
- 3 Does it survive finite width?

Low-frequency structure is learned before high-frequency detail.

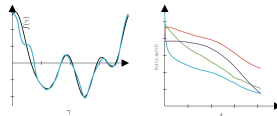
$\cos(1\gamma)$ $\cos(2\gamma)$ $\sin(3\gamma)$ $\cos(4\gamma)$



One-hidden-layer MLP



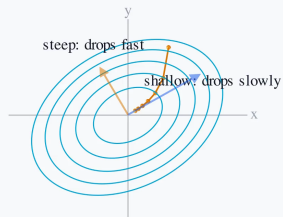
Two-hidden-layer MLP



Attention Block

The error is a landscape

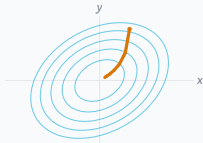
Seen from above, the contour rings are the bowl's height; training rolls toward the bottom.



Each direction has its own rate

Training makes the error small by walking downhill, along the steepest drop.

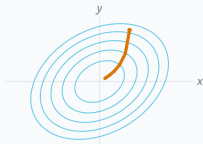
recall the error: $\frac{1}{n} \sum_i (f(x_i) - f^*(x_i))^2$



In your x, y axes the route bends: the two directions are coupled.

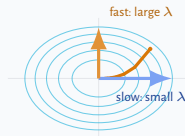
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$\Rightarrow$
the bowl's own
axes

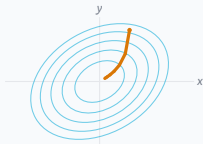


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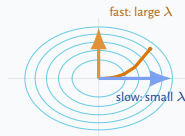
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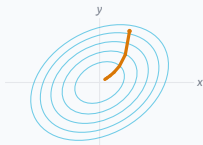
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Along one direction, the error falls in proportion to how much is left:

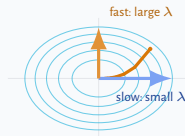
$$\dot{\alpha}(t) = -\lambda \alpha(t)$$

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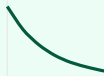
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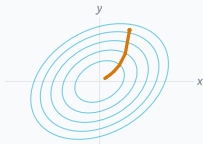
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like cooling or half-life; λ is the rate.

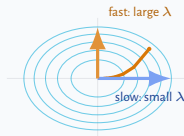


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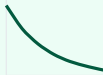
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Eigenvector: a direction the dynamics keeps separate, never mixing it with others. **Eigenvalue λ :** its rate, large is fast, small is slow.

For a neural network, what are these directions and rates?

The rate becomes an operator: the tangent kernel

The same law as before, with the single rate replaced by an operator.

One direction

$$\dot{\alpha}(t) = -\lambda \alpha(t)$$

The error falls at rate λ , a single number.

A whole function

$$\partial_t r_t = -T_t r_t$$

The rate becomes an operator T_t , the network's tangent kernel (NTK).

If T_t were fixed, diagonalizing it would split this into one copy of the scalar law per eigenfunction, $\dot{\alpha}_k = -\lambda_k \alpha_k$. The eigenfunctions are the directions; the eigenvalues are the rates.

So the question becomes: what is the spectrum of T ?

Why we start from an idealized model

To read off a fixed set of directions and rates, T must be one fixed operator. Two things are in the way.

Time-dependent

T_t changes as the network trains, so its directions and rates drift.

Freeze it to get a constant operator.

Finite scale

At finite data T depends on the sampled points; at finite width it is random.

Take the continuum and infinite width for one deterministic operator.

Strategy: build the clean reference, then put each idealization back.

Reference



Finite data



Finite width



Evolving

Setting: regression on S^1

S^1 : functions on the circle have a natural Fourier decomposition.

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Domain + target

$$x(\varphi) = (\cos \varphi, \sin \varphi) \in S^1$$

$$y(\varphi) = \sum_k c_k \phi_k(\varphi)$$

Fourier basis: $1, \cos(k\varphi), \sin(k\varphi)$.

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Network

$$f_m(x; \theta) = \frac{1}{\sqrt{m}} \sum_{i=1}^m a_i \sigma(z_i)$$
$$z_i = w_i^\top x + b_i$$

$$\sigma(u) = \max\{0, u\}$$

$$a_i, w_i \sim \mathcal{N}(0, I), \quad b_i = 0$$

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Training dynamics

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$$r_t = f_t - y$$

$T_t^{(m)}$ is the tangent operator.

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Key question: what governs the frequency-dependent decay of $\alpha_k(t) = \langle r_t, \phi_k \rangle$?

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Explicit eigenvalues

$$\lambda_k = \begin{cases} -\frac{1}{4} + \frac{2}{\pi^2}, & k = 0, \\ -\frac{1}{4} + \frac{1}{\pi^2}, & k = 1, \\ \frac{k^2 \pi^2}{k^2 + 3}, & k \geq 2 \text{ even}, \\ \frac{1}{\pi^2 k^2}, & k \geq 3 \text{ odd}. \end{cases}$$

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- ▶ Small k have larger eigenvalues, so low frequencies decay faster.
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Reference model: exact Fourier dynamics at infinite width and in the continuum.

Does this spectral picture survive outside the ideal limit?

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The ideal model is useful only if we understand its perturbations.

Perturbation I

Finite samples

$$\mu \rightsquigarrow \frac{1}{n} \sum_{i=1}^n \delta_{\varphi_i}$$

Replace the continuous measure by n random points on S^1 .

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Frozen finite width

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Question

Does the finite-width tangent operator preserve the Fourier eigenspaces and learning rates?

Both perturbations break exact diagonalization; the goal is to quantify the finite-scale error.

Main results: explicit finite-scale error bounds

On a fixed low-frequency Fourier subspace $\mathcal{H}_K$ with $d = 2K + 1$, both perturbations are controlled.

Finite samples

On the Fourier block $\mathcal{H}_K$:

$$A_{ij} = \frac{1}{n} T_{\infty}(\theta_i, \theta_j)$$

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$$\Lambda_K = P_K T_{\infty} P_K$$

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Finite samples

Thm. 5.5 + Cor. 5.6

With probability at least $1 - \delta$:

$$\begin{aligned}\|G - I\|_{\max} &\leq c_1 \sqrt{\frac{\log(d/\delta)}{n}}, \\ \|H - \Lambda^{(n)}\|_{\max} &\leq c_2 \sqrt{\frac{\log(d/\delta)}{n}}, \\ \|A\Phi - \Phi\Lambda^{(n)}\|_{\max} &\leq c_3 \sqrt{\frac{\log(nd/\delta)}{n}}.\end{aligned}$$

Frozen finite width

Thm. 5.7

With probability at least $1 - \delta$:

$$\mathbb{E} B_m^{(K)} = \Lambda_K$$

$$\|B_m^{(K)} - \Lambda_K\|_{\max} \leq c_4 \sqrt{\frac{\log(d/\delta)}{m}}$$

Summary: Fourier structure survives finite sampling and frozen finite width with explicit high-probability control.

A finite-width effect not explained by the frozen NTK

After controlling the frozen operator, the next question is whether training changes the operator enough to matter.

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[Figure] Training loss: frozen vs. evolving kernel — width $m = 128$

width $m = 128$: evolving kernel reaches much lower loss

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[Figure] Training loss: frozen vs. evolving kernel — width $m = 128$

width $m = 128$: evolving kernel reaches much lower loss

[Figure] Training loss: frozen vs. evolving kernel — width $m = 1024$

width $m = 1024$: frozen picture becomes competitive

- ▶ At small width, the evolving network is not well described by simply freezing the NTK at initialization.
- ▶ As width grows, the difference between frozen and evolving dynamics becomes much smaller.
- ▶ This creates the main tension: finite width can help by changing the kernel, but it also breaks the clean spectral reference model.

What changes when the operator moves?

The evolving-kernel regime keeps the same residual equation, but the operator is now time-dependent.

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the learning operator changes during training

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the learning operator changes during training

Empirical picture

- ▶ Small widths show substantial kernel movement.
- ▶ Low-frequency spectral strength can increase.
- ▶ This can accelerate training loss decay.
- ▶ But eigenspaces can drift away from the Fourier reference.

Theoretical gap

The finite-sample and frozen-width regimes are controlled by concentration.

For the evolving kernel, one would need non-asymptotic control of $T_t^{(m)}$ — $T_0^{(m)}$ over training time.

Thesis takeaway: the frozen NTK gives a controlled spectral reference; finite-width training shows where that reference becomes insufficient.

Future direction: mathematical structure in function space as an explanation for training and generalization.